NIR2FPGA: Compiling Spiking Neural Networks into RTL

Michail Rontionov

University of Southampton

30th April 2026

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H_1

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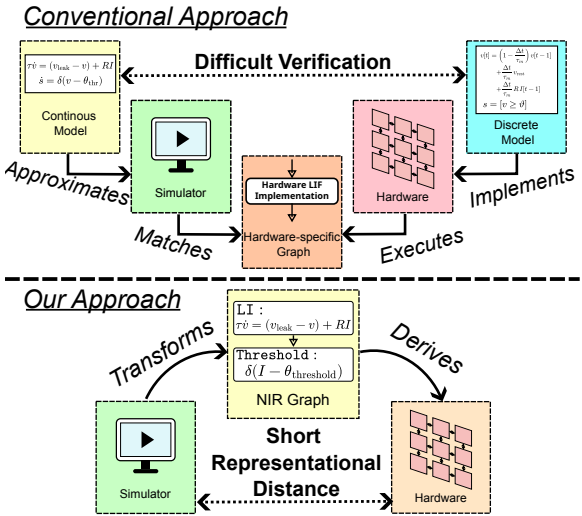

NIR2FPGA: Generating Dataflow Accelerators from the Neuromorphic Intermediate Representation

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David Thomas[1], Charlotte Frenkel[4], Jens Egholm Pedersen[5‡]
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[3]ETH Zürich, Zürich, Switzerland [4]Delft University of Technology, Netherlands
[5]Technical University of Denmark
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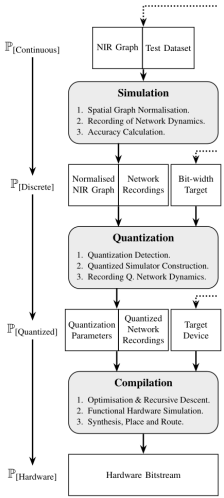
What does it do?

- Example scenario: "I have an SNN with the *CuBa-LIF* neuron model, implemented in a *NIR-supported simulator*. I would like to evaluate its edge performance in hardware".



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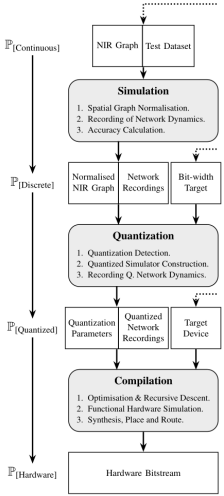

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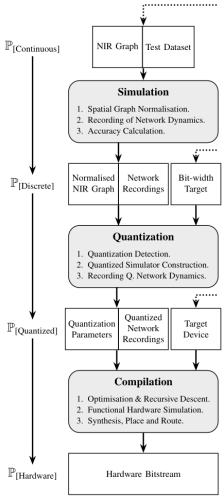

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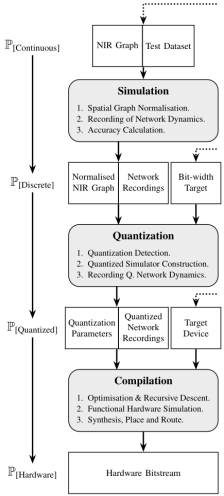

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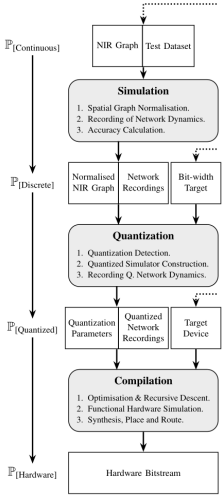

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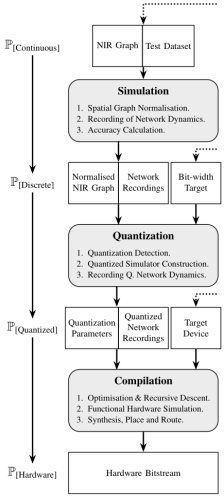

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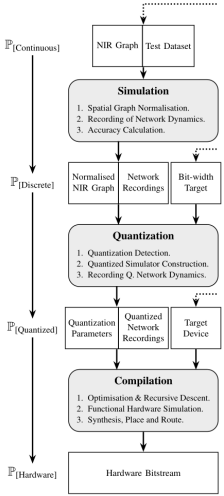

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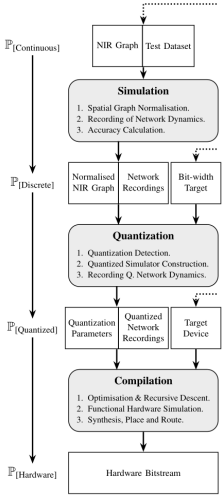
- Outputs:
  - Hardware Bitstream according to inputs.



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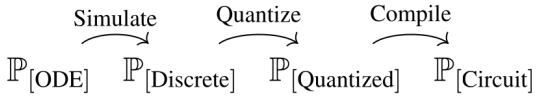
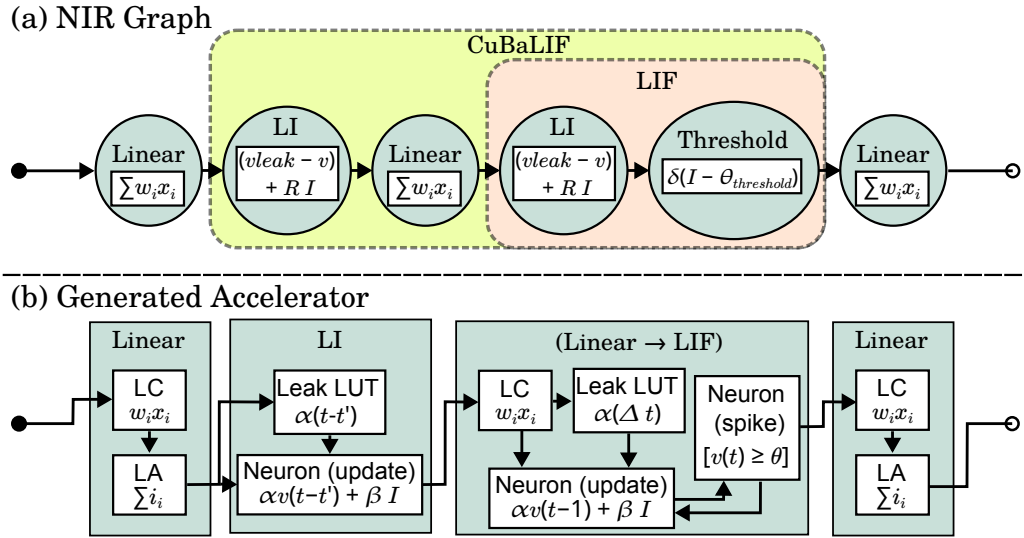
We model this as three separate unidirectional transformations



Example Generation



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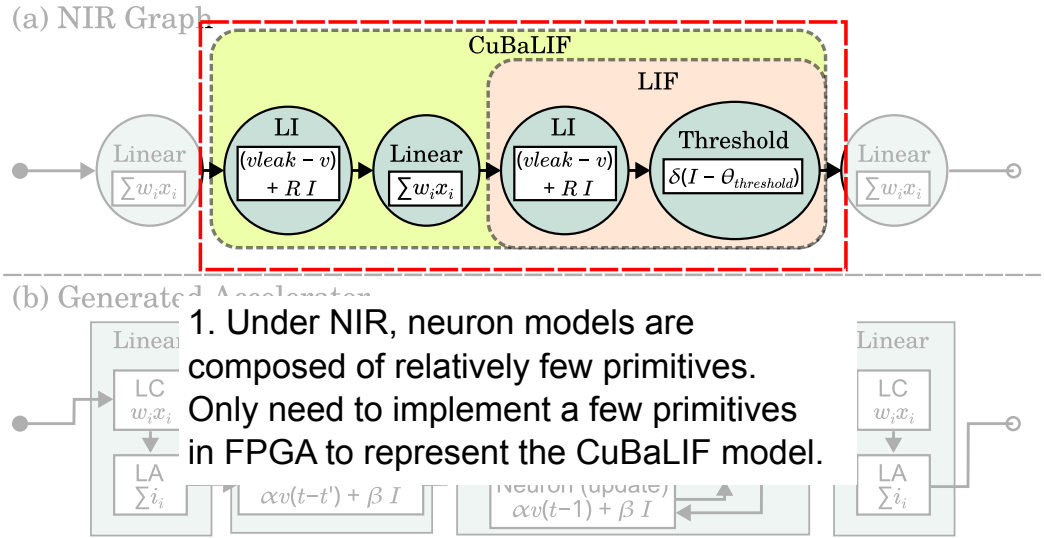

1. Under NIR, neuron models are composed of relatively few primitives.
Only need to implement a few primitives in FPGA to represent the CuBaLIF model.

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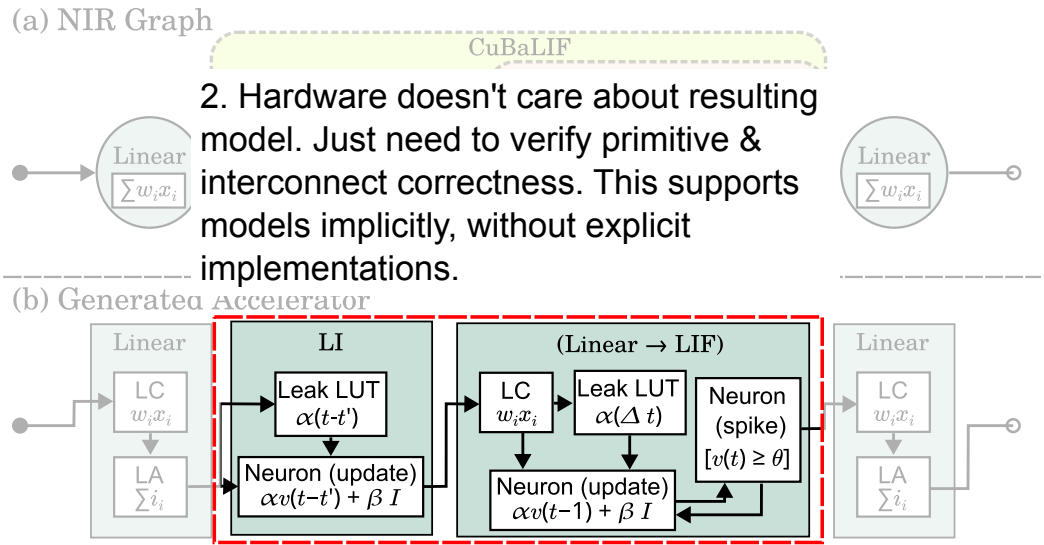
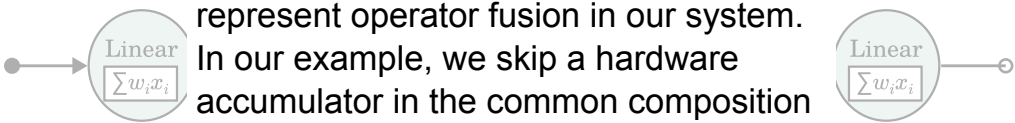


Example Generation

(a) NIR Graph

CuBaLIF

3. Assuming NIR, we can effectively represent operator fusion in our system. In our example, we skip a hardware accumulator in the common composition Linear -> LIF.



(b) Generated Accelerator